FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 3.2: LINEAR MOTION	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation task. You will write a structured, evidence-based answer to the driving question using everything you have learned across all 8 lessons on Linear Motion. Read the driving question carefully, then complete each section below in your own words. Use correct scientific vocabulary, equations, and real examples from the mkokoteni phenomenon. Your goal is to show that you truly understand why the cart accelerates, and that you can use mathematics to predict its motion — not just describe it. Write in full sentences and show all working where calculations are required.

Section 1: Define the Motion – What Was the Mkokoteni Actually Doing?	
Describe the motion of the mkokoteni as it rolled down the Limuru escarpment. Use precise scientific vocabulary in your answer. In your response, define and distinguish between: displacement, velocity, speed, and acceleration. Explain whether the cart's acceleration was uniform or non-uniform, and justify your answer. State the cart's initial condition (it started from rest) and explain what 'from rest' means mathematically.	When the mkokoteni was released from rest at the top of the Limuru escarpment, it began a journey of continuously changing motion. 'From rest' means its initial velocity was zero ($u = 0$ m/s). As it rolled downhill, its displacement — the straight-line distance from its starting point in a specific direction (downhill) — increased with every passing second. Its speed (how fast it moved, regardless of direction) and velocity (speed in a specific direction — down the slope) both increased continuously. This increase in velocity over time is called acceleration. Because gravity pulled the cart downhill with a roughly constant force along the slope, and because we assume friction is negligible, the acceleration was approximately uniform — meaning the cart gained the same amount of velocity every second. This is the key reason why the cart kept getting faster and faster without anyone pushing it: gravity provided a continuous, unrelenting force.

Section 2: The Invisible Force – Why Did the Cart Keep Accelerating?	
Explain the cause of the cart's	The mkokoteni accelerated because of gravity — the same invisible force that pulls every object toward the centre of the Earth.

acceleration. Why did it keep getting faster even though no person or engine was pushing it? In your answer: (a) identify the force responsible, (b) explain what gravitational acceleration (g) is and give its approximate value on Earth, (c) connect this to the concept of free fall, and (d) explain how the steepness of the Limuru escarpment affects the acceleration. Use the KICD inquiry question — 'How is free fall motion important in real life?' — to frame your answer.	Even though no one was pushing the cart, gravity continuously acted on it, doing work on it every metre it travelled downhill. Near the Earth's surface, gravity gives every freely falling object an acceleration of approximately $g = 10 \text{ m/s}^2$ (or more precisely, 9.8 m/s^2). This means that in every second of free fall, an object gains 10 m/s of velocity. Free fall is the purest form of this: an object falling under gravity alone, with no other forces acting. The mkokoteni was not in true free fall (it was rolling on a road, not falling through air), but it was a real-life approximation — a practical example of gravitational acceleration at work. The steepness of the Limuru escarpment matters because a steeper slope means a greater component of gravitational force acts along the direction of motion, so the effective acceleration is larger. This is why escarpments like Limuru are so dangerous — the slope is steep enough that carts and vehicles can reach deadly speeds in just a few seconds. This answers the inquiry question: free fall and gravitational acceleration are critically important in real life because they govern how quickly runaway objects, falling people, or collapsing structures gain speed.

Section 3: The Mathematics of Motion – Predicting Speed and Distance	
Use the equations of linear motion (equations of uniformly accelerated motion) to answer the following prediction problem: **The mkokoteni starts from rest ($u = 0$ m/s) and rolls down the escarpment with a uniform acceleration of 4 m/s^2.** **(a) What is the cart's velocity after 6 seconds? (Use $v = u + at$)** **(b) How far has the cart travelled after 6 seconds? (Use $s = ut + ½at^2$)** **(c) What is the cart's velocity after it has travelled 200 m? (Use $v^2 = u^2 + 2as$)** **Show all your working clearly, including the formula used, substitution of values, and the final answer with correct units.**	We are given: $u = 0$ m/s, $a = 4 \text{ m/s}^2$, $t = 6$ s, and later $s = 200$ m. (a) Velocity after 6 seconds: Formula: $v = u + at$ Substituting: $v = 0 + (4)(6)$ $v = 24$ m/s The cart is travelling at 24 m/s (about 86 km/h) after just 6 seconds — faster than most vehicles on a Nairobi road. (b) Distance travelled after 6 seconds: Formula: $s = ut + ½at^2$ Substituting: $s = (0)(6) + ½(4)(6^2)$ $s = 0 + ½(4)(36)$ $s = 72$ m In just 6 seconds, the cart has covered 72 metres — nearly the length of a standard athletics straight at a Kenyan secondary school. (c) Velocity after travelling 200 m: Formula: $v^2 = u^2 + 2as$ Substituting: $v^2 = 0^2 + 2(4)(200)$ $v^2 = 1600$ $v = \sqrt{1600} = 40$ m/s After 200 m, the cart is moving at 40 m/s (144 km/h) — a catastrophically dangerous speed. These numbers show that the mathematics of uniformly accelerated motion is not just abstract: it lets us predict exactly how fast and how far the mkokoteni would

After your calculations, explain in words what these numbers tell us about the danger of the situation.	travel, and it confirms why the crash would have been so devastating.

Section 4: Mitigating the Danger – Applying Understanding to Save Lives	
The second KICD inquiry question asks: 'How can the dangers of free fall and runaway acceleration in our day-to-day life be mitigated?' Using what you know about acceleration, velocity, and the equations of motion, suggest and explain at least THREE specific, practical measures that could prevent a mkokoteni accident on the Limuru escarpment or reduce its severity. For each measure, explain the physics behind why it would work — do not just list ideas without explanation.	Understanding the physics of acceleration allows us to design real solutions to prevent tragedies like the mkokoteni accident. 1. Runaway truck escape ramps (gravel arrester beds): On steep escarpments like Limuru, engineers can build side ramps filled with deep gravel. When a runaway vehicle or cart enters the gravel, the friction force dramatically increases, producing a large deceleration (negative acceleration) that rapidly reduces velocity to zero. The physics: a large opposing force means a large deceleration, so the time and distance needed to stop are greatly reduced ($v^2 = u^2 + 2as$ — increasing the deceleration 'a' in the negative direction reduces the stopping distance 's'). 2. Wheel chocks and braking mechanisms on mkokoteni: A simple wooden or metal chock placed behind a wheel before parking on a slope prevents the cart from moving at all (u remains 0). This is the simplest intervention: prevent the initial release so acceleration never begins. Drivers and cart owners should be trained to always use chocks on slopes. 3. Reducing the load on steep slopes: A heavier cart experiences a larger gravitational force, but since $F = ma$, for the same slope the acceleration is the same regardless of mass (gravity accelerates all masses equally). However, a heavier cart is harder to stop and causes more damage on impact. Breaking large loads into smaller trips reduces the momentum ($p = mv$) and the kinetic energy ($KE = \frac{1}{2}mv^2$) at any given speed, reducing crash severity. These solutions show that physics education is not just academic — it directly saves lives on Kenyan roads.

Section 5: Revisiting the Driving Question – Your Complete Answer	
Now bring everything together. Write a single, well-organised paragraph (minimum 8 sentences) that directly and fully answers the driving question: 'Why does a runaway mkokoteni on the Limuru escarpment keep getting faster — and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?' Your answer must: (a) explain the cause of acceleration, (b)	The mkokoteni on the Limuru escarpment keeps getting faster because gravity — an invisible but constant force — continuously accelerates it downhill from the moment it is released from rest. This is directly related to gravitational acceleration, $g \approx 10$ m/s², which causes every object near Earth's surface to gain speed at a rate of 10 m/s every second when falling freely, and a proportional acceleration when rolling down a slope. Because the cart starts from rest ($u = 0$ m/s) and the gravitational force acts uninterrupted, the acceleration is uniform, meaning the cart gains equal amounts of velocity in equal time intervals — which is why it becomes dangerously fast so quickly. We can use the equations of uniformly accelerated motion to predict its motion with precision: using $v = u + at$, we can calculate the cart's velocity at any moment in time, and using $s = ut + \frac{1}{2}at^2$, we can find exactly how far it has travelled. For example, with an acceleration of 4 m/s², the cart reaches 24 m/s after just 6 seconds and covers 72 metres — and using $v^2 = u^2 + 2as$, we find it reaches 40 m/s (144 km/h) after only 200 metres. This is the power of the equations of motion: they transform a frightening, chaotic event into something we can analyse, predict, and prepare for. The real-world implication is profound — road engineers, county governments, and cart operators in areas like Limuru, Kijabe, and the Rift Valley can use exactly these calculations to determine where to place escape ramps, warning signs, and braking systems, potentially saving the lives of drivers, traders, and pedestrians on Kenya's steep escarpment roads.

reference at least two equations of motion by name or formula, (c) mention at least one calculated result, (d) connect to free fall and the value of g, and (e) state one real-world implication for safety in Kenya.	

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding of Acceleration & Gravity	Accurately and thoroughly explains gravitational acceleration ($g \approx 10\ m/s^2$), uniform acceleration, and free fall. Clearly connects all concepts to the mkokoteni phenomenon with precise scientific vocabulary used consistently and correctly throughout.	Correctly explains gravitational acceleration and uniform acceleration and connects them to the mkokoteni. Minor gaps in use of scientific vocabulary or depth of explanation, but overall understanding is clear.	Shows partial understanding of acceleration or gravity but confuses key terms (e.g., speed vs. velocity, acceleration vs. velocity). Connection to the mkokoteni phenomenon is weak or missing.
Application of Equations of Motion	Correctly selects and applies all three equations of motion ($v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$) with full working shown — formula stated, values substituted, answer given with correct units. All three calculations are accurate.	Correctly applies at least two of the three equations with working shown. One minor arithmetic or unit error may be present, but the method is sound and the student demonstrates understanding of when and why each equation is used.	Attempts to use equations but makes significant errors in formula selection, substitution, or calculation. Working is incomplete or units are missing. Fewer than two equations are applied correctly.
Use of Evidence and Scientific Vocabulary	Consistently uses precise scientific terminology (displacement, velocity, acceleration, uniform acceleration, free fall, initial velocity, deceleration) accurately throughout all sections. All claims are supported with evidence from calculations or the phenomenon.	Uses most scientific terms correctly with only occasional imprecision. Most claims are supported by evidence. The student demonstrates a working scientific vocabulary appropriate for Grade 10.	Uses everyday language in place of scientific terms, or misuses key vocabulary (e.g., uses 'speed' and 'velocity' interchangeably without distinction). Claims are often made without supporting evidence or calculation.
Real-World Application & Safety Mitigation	Proposes at least three specific, realistic safety measures for Kenyan escarpment contexts. Each measure is fully explained using correct physics principles (e.g., friction increases deceleration, reducing s needed to stop). Demonstrates clear transfer of learning to real-life situations.	Proposes at least two safety measures with a reasonable physics explanation for each. The connection to Kenyan contexts is present. One measure may lack a full physics justification.	Lists safety ideas without explaining the underlying physics, or the measures proposed are vague and not connected to the concepts of acceleration, force, or velocity studied in the lessons.
Coherence of the Driving Question Response	The final section paragraph directly, fully, and eloquently answers the driving question. All five required elements are present (cause of acceleration, two	The final paragraph answers the driving question and includes most of the five required elements (at least four). The response is logical and mostly well-	The final paragraph only partially answers the driving question. Fewer than three of the five required elements are present, or the response is a list of

	equations, one calculated result, link to free fall/g, one safety implication). The response reads as a unified, logical argument.	organised, though one element may be underdeveloped.	disconnected facts rather than a coherent, evidence-based argument.